Testing, Validation & Security Phase

Making sure the solution works, is accurate, and is secure

1. What is this phase?

After the solution is built, it must be **checked before it goes live**. This phase confirms three things: the solution **works as expected**, it **meets every functional requirement**, and it follows **basic security and access control** practices. It covers the complete automation from start to finish and makes sure ticket data is correct and protected.

2. Breaking down the objective

Phrase	Meaning
Works as expected	Every feature behaves the way it was designed, with no unexpected errors.
Meets functional requirements	Each requirement in the project scope is implemented and can be shown working.
Basic security and access control	Users only see and do what their role allows.
End-to-end automation	The full flow is tested from the first trigger (for example, ticket creation) to the final result (for example, assignment, notification, closure).
Data accuracy	Fields, values, states and notifications are correct at every step, with no missing or wrong data.
Only authorized users	Unauthorized users cannot view, edit or delete ticket data.

3. Part A: Functional and end-to-end testing

- **Prepare test cases:** list each requirement with the input, the steps and the expected result.
- **Run the full flow:** create a ticket and watch the automation (rules, flows, assignments, notifications) run from start to end.
- **Check positive and negative cases:** valid data should work, and invalid or missing data should be handled properly with clear messages.
- **Try edge cases:** empty fields, duplicate tickets, different priorities, and different categories.
- **Record results:** mark each test as Pass or Fail, and fix and retest failures.

4. Part B: Data validation

- Confirm that field values are saved correctly (short description, category, priority, assignment group, state).
- Check that auto-populated and calculated fields show the right values.
- Verify that notifications and records go to the right people, with correct content.
- Make sure no duplicate or incorrect records are created by the automation.

- Check the audit trail or activity log so changes can be traced.

5. Part C: Security and access control

Area	What to check
Roles	Each user has only the roles they need (least privilege), for example a requester, a fulfiller and an admin.
ACLs (Access Control Rules)	Read, write, create and delete rules on the ticket table and its fields work as designed.
Test with different users	Log in (or impersonate) as different roles and confirm allowed actions succeed and blocked actions are denied.
Sensitive fields	Restricted fields are hidden or read-only for users who should not see or change them.
No hard-coded secrets	Passwords, tokens and API keys are stored securely, not inside scripts.
Input handling	Scripts validate user input and avoid unsafe practices.

6. Sample test case format

ID	Scenario	Expected result	Actual	Status
TC01	Create ticket with all valid data	Ticket created, auto-assigned, notification sent		Pass / Fail
TC02	Create ticket with mandatory field empty	Error shown, ticket not saved		Pass / Fail
TC03	Normal user tries to edit another user's ticket	Access denied		Pass / Fail
TC04	Fulfiller updates an assigned ticket	Update allowed and logged		Pass / Fail
TC05	Close ticket and check state and data	State updated, data accurate		Pass / Fail

7. Exit checklist

Item	Done?
All functional requirements tested	■
End-to-end automation verified	■
Data accuracy confirmed	■
Roles and ACLs tested with different users	■
Unauthorized access is blocked	■

Item	Done?
Defects fixed and retested	■
Test results documented	■

Summary: This phase proves the solution works from start to finish, keeps data correct, and lets only authorized users access or change ticket data. Once it passes, the solution is ready for deployment.